

Does digital dividend matter in China's green low-carbon development: Environmental impact assessment of the big data comprehensive pilot zones policy

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ABSTRACT

Currently, data is increasingly becoming a crucial production factor and strategic resource for economic development. However, there is no consensus on whether data plays a significant role in promoting city green development. This paper examines the effect and the mechanism of big data on carbon emission reduction based on a quasi-natural experiment of Big Data Comprehensive Pilot Zones (BDCPZ) in China, using a difference-in-difference approach and combining panel data from 261 cities in China from 2010 to 2019. The study found that: (1) The policy of BDCPZ significantly reduced carbon emissions, which confirms the critical importance of the digital dividend in China's green development process. (2) We find the emission reduction effect of BDCPZ is achieved in three main ways: promoting industrial upgrading, improving energy use efficiency and increasing green TFP in manufacturing. (3) Further analysis reveals that the carbon emission reduction effect of BDCPZ varies depending on city population size, levels of manufacturing development, resource endowments, and pilot cities with low carbon related policies. This paper empirically confirms the digital dividend in city green transformation, provides evidence and policy guidance for the broad application of the digital economy in the field of green and low-carbon development, and enriches research in development economics and environmental economics. It helps to provide novel policy insights for the government to implement proactive policies of carbon reduction, and to build a diversified carbon reduction governance system to successfully achieve “carbon peak” and “carbon neutral” targets.

1. Introduction

Global warming, a serious challenge for humanity, has critically affected sustainable economic and social development. China has achieved remarkable economic achievements over the past four decades of reform and opening-up. However, as the world's largest emitter and energy consumer, China's energy conservation and emission reduction efforts have an important role to play in mitigating global warming (Cheng et al., 2020). At the same time, as the world's largest developing country, China's experience in contributing to the carbon reduction path can provide some inspiration for other developing countries on their path to green development.

The government of China has made a series of efforts and has made certain achievements on carbon reduction. According to the statistics, China's CO₂ emissions per unit of GDP will drop by 18.7% in 2020

compared with 2015 and by 48.4% compared with 2005, reversing the rapid growth in CO₂ emissions. In 2021, the country will reduce its national carbon intensity by 3.8% compared to 2020. However, against the backdrop of the new economic normal, China's carbon emission reduction task remains daunting, given the reality that manufacturing products have long been as “cheap China”, the high energy consumption and low added value of products, the relatively large share of coal consumption, and the still high energy consumption per unit of GDP (Qian et al., 2019). In this context, how to successfully achieve the “Double Carbon” goals has become a major issue for society and academia. From a macro perspective, the achievement of the carbon reduction target is important for China to promote a green society. At the micro level, the achievement of the “Double Carbon” target is also essential for increasing people's well-being.

Currently, the research topic of “Double Carbon” has attracted the

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attention of academic and policy authorities. In the existing studies, it was based primarily on economic growth (Narayan et al., 2016; Destek et al., 2020), industrial structure (Zhang et al., 2020), international trade (Wang et al., 2022b; Hao and Liu, 2014), environmental regulation (Neves et al., 2020; Du and Li, 2020; Huang and Yi, 2023; Lin and Zhang, 2023), green technology Innovation (Dong et al., 2022; Du et al., 2019; Lin and Ma, 2022) and others to discuss the deeper causes of carbon emissions. Other literature has focused on low-carbon city pilots (Huo et al., 2022; Zhang et al., 2022a), carbon emissions trading pilots (Shi et al., 2022; Wang et al., 2019a), smart city construction (Guo et al., 2022; Fan et al., 2021) and other pilot policies to explore paths of carbon emission reduction.

It is worth noting that China has made crucial progress in data construction and digital economy. Therefore, an in-depth study of China's digital emission reduction pathway is needed to provide lessons for other countries on the way to green development through digital tools. Nowadays, digital technology has penetrated into all aspects of production and life, and has gradually become an important tool to promote green development (Li et al., 2021; Chen et al., 2022; Zhang et al., 2022c; Yi et al., 2022). It is also argued that the application of Information Communication Technology (ICT) technologies and digital economy have not only positive but also some negative effects on the environment and greenhouse gas emissions (Zhou et al., 2019).

The existing literature has focused on the relationship between the digitalization of enterprises (Cao et al., 2023; Chen et al., 2023; Shang et al., 2023), the level of the regional digital economy (Zhang et al., 2022b; Yi et al., 2022) and green development, while some scholars have explored the path of green transformation based on quasi-natural experiments such as smart city (Guo et al., 2022; Fan et al., 2021) and broadband China (Wang et al., 2022a; Hong et al., 2023). However, the existing literature does not focus enough on the power of data in achieving carbon reduction and green development. In particular, the actual effect and mechanism of building comprehensive pilot zones for big data on carbon reduction has not been fully demonstrated. This paper measures carbon emissions of 261 cities in China from 2010 to 2019 based on the energy consumption, followed by the "Big Data Comprehensive Pilot Zones" (BDCPZ) approved by the Chinese government in 2016 to explore the carbon emission reduction effects and specific mechanisms of Big Data. We attempt to answer the following questions with a rigorous empirical analysis: Does the construction of the BDCPZ in China help reduce carbon emissions in cities? What is the mechanism? Are there heterogeneities in different conditions?

The main marginal contributions of our findings are: first, we are situated within a growing body of literature studying the relationship between digital economy and green development. However, the existing research has not evaluated the impact of BDCPZ construction on green and low-carbon development, this paper fills the research gap in related fields. We assess the real effect of big data construction on carbon emission reduction by taking the establishment of China's BDCPZ as a quasi-natural experiment for the first time, which confirms the existence of digital dividend in the process of city green transformation, significantly expands policy toolbox of decarbonization technologies, and enriches relevant research in development economics and environmental economics. Second, in terms of policy implications, our research has fully revealed the process and impact mechanism of big data on carbon emission reduction, and examines the applicability of big data construction in different conditions, thus opening the black box of "digital dividend", which not only provides scientific basis for the application of big data to environmental governance, but also provides novel policy insights for the government to deploy emission reduction strategies according to different scenarios. Although this paper refers to the empirical evidence of China, the relevant experience in the construction of big data can be extended to other developing countries, especially to provide some inspiration for other countries on how to release the carbon emission reduction dividend of big data.

The rest of this paper is organized as follows. Section 2 introduces the

background of BDCPZ policy and elaborate the influencing mechanism of on carbon reduction. In Section 3, the research design is presented. Section 4 reports the empirical results and makes corresponding analysis. Section 5 concludes and puts forward policy implications.

2. Policy background and mechanism elaboration

2.1. Policy background for the construction of BDCPZ

Data has become an essential factor of production for economic and social development in this age, helping to improve the quality and efficiency of labour, technology, knowledge, capital and other factors (Ghasemaghaei and Calic, 2019). On 25 February 2016, the Chinese government issued a letter of approval, formally agreeing to build big data comprehensive pilot zone in Guizhou. Since then, the construction of the BDCPZ has kicked off. In October of the same year, the second batch of provinces approved was released (See Fig. 1.). So far, the Chinese government has set up a total of eight big data pilot zones, which will jointly lead the development of big data industry in "four major sectors" (eastern, central, western and northeastern), achieve data sharing, collaborative development within the region, accelerate industrial transformation within the region, and further drive the application and development of big data industry in the surrounding areas.

It should be noted that since cities in the same province of the pilot zone are significantly different in terms of city size and economic development level, if all cities in the same province are included in the treatment group indiscriminately, the assessment results may be biased from the actual situation. Therefore, the selection of cities for the treatment groups in this paper has been made with great care. We further screen and classify cities by referring to a large number of policy documents and development plans of various regions.

Specifically, the Beijing-Tianjin-Hebei cross-regional pilot zone was clearly proposed at the early stage of construction, with Beijing and Tianjin as integrated support, Zhangjiakou in Hebei Province as a new energy demonstration area for big data, Langfang as a demonstration area for logistics and financial remote sensing big data, Chengde as a tourism demonstration area for big data. Qinhuangdao as a demonstration area for health big data, and Shijiazhuang as a demonstration area for applications of big data. Therefore, Beijing, Tianjin, Zhangjiakou, Langfang, Chengde, Qinhuangdao and Shijiazhuang are included in the treatment group. In Henan, during the construction of the big data pilot zone, the spatial distribution pattern of Zhengzhou as the core, Luoyang as the important support and other cities as the main nodes is clearly proposed, therefore all cities in Henan are included in the treatment group. Considering that Guizhou Province and Inner Mongolia Autonomous Region have a poor foundation for digital economy development, and that most of their big data construction plans issued around 2016 mention Guizhou, Anshun and Hohhot, which are included as the treatment group, and other cities in the two provinces are included as the control group. We also used Shanghai, Chongqing, Shenyang and all cities in Pearl River Delta (PRD) as the treatment groups.

2.2. Mechanism elaboration and research hypotheses

2.2.1. The impact of BDCPZ on carbon emission

The construction of BDCPZ through the deep cultivation of big data industry, vigorously develop the digital economy, and give full play to ecological advantages to continuously enable the green development of the city, which has a direct impact on urban energy conservation and emission reduction. Firstly, according to its own characteristics, big data promotes economic transformation and upgrading, demand structure optimization, quality and efficiency improvement. Therefore, the construction of BDCPZ itself has green properties. Secondly, big data has the interconnection of everything. At a micro level, the convenience of digital technological advances has significantly reduced the search,

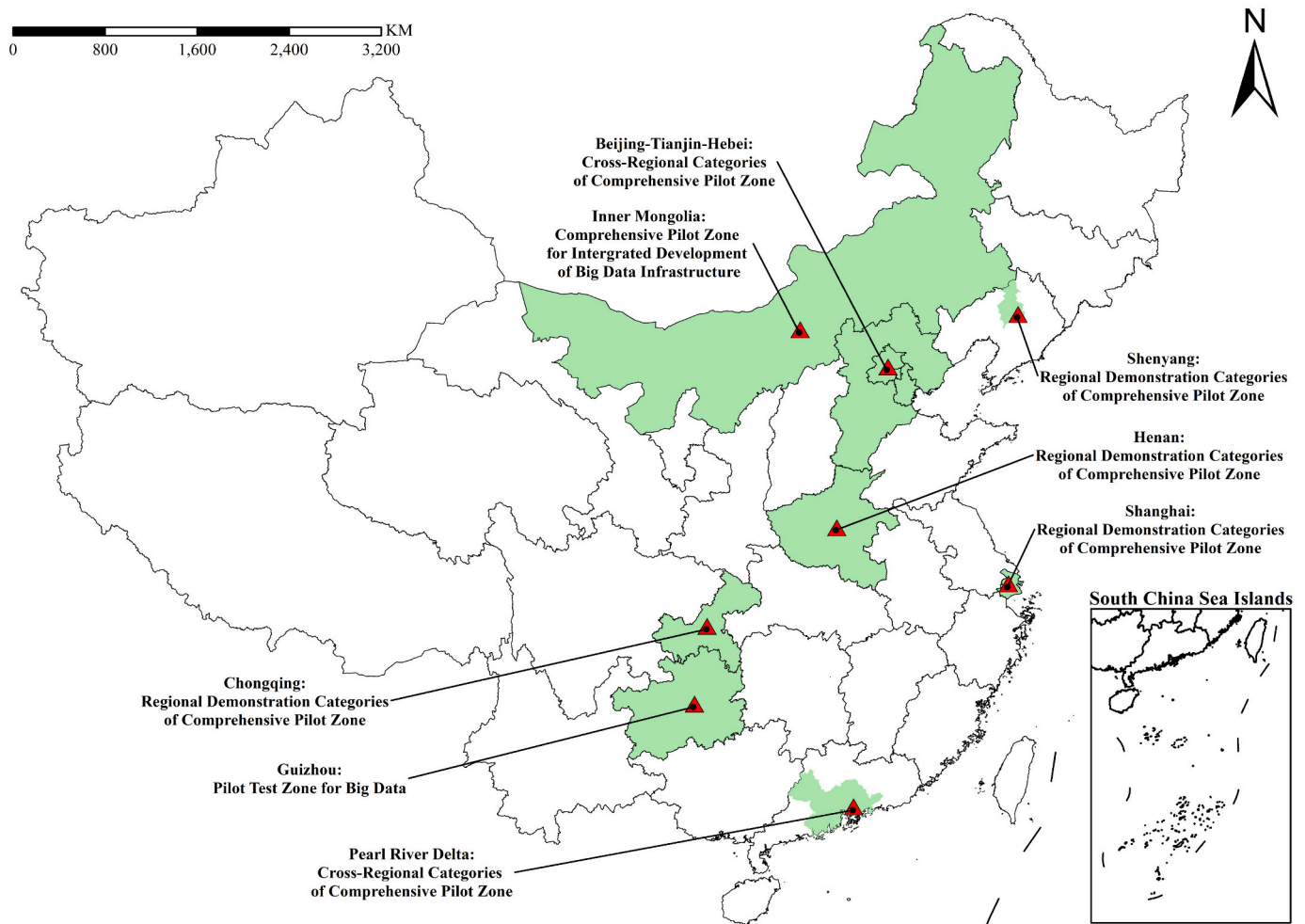


Fig. 1. The geographical of BDCPZ.

replication, transportation and verification costs for firms conducting complex economic activities over long distances (Goldfarb and Tucker, 2019), effectively increasing the synergies between factors of production. From a macro perspective, the application of digital technology has the ability to integrate information resources across regions, thus enhancing cross-regional collaborative innovation (Liu et al., 2020), improving factor utilization efficiency and flow efficiency and reducing redundancy. Thirdly, the application of data infrastructure and digital technology has facilitated the “virtualization” of economic operations (Pradhan et al., 2020), altering traditional business models and inducing profound transformations across all links in the industrial chain. This has reduced dependence on heavy industry (Horner et al., 2016) and decreased carbon emissions (Yang et al., 2022). Based on the above analysis, **Hypothesis 1** is proposed in this paper.

Hypothesis 1. The construction of BDCPZ can effectively reduce the level of carbon emissions in cities.

2.2.2. The mechanism of BDCPZ on carbon emission

The adjustment of industrial and energy structure are the two main factors affecting carbon emission intensity (Chen, 2011), while green technological progress, has shown a “technological dividend” for low-carbon transition development (Shao et al., 2022). Therefore, this paper explores specific mechanisms for building comprehensive pilot zones for big data to reduce the level of city carbon emissions from three perspectives: industrial structural upgrading, energy efficiency

improvement, and green TFP improvement in manufacturing.

2.2.2.1. Industrial structure effect. The construction of BDCPZ could reduce carbon emissions levels by optimizing the city's industrial structure. First, big data and the real economy have been deeply integrated, and the digital foundation for high-quality development has been continuously consolidated. Compared with the traditional industrial development model, the construction of BDCPZ has substantially improved the efficiency of the use and flow of data as a factor of production within and between the various industries in the pilot zone. With the network externality of the digital economy, the construction of pilot continues to spill over digital technologies to traditional industries, promoting digitization, intelligence, low energy consumption and low emission transformation and upgrading of traditional industries under the aegis of big data. Second, the wide application of digital technology in pilot zones will trigger industrial transformation and accelerate the development of new industries, new business forms and new models. Data as a factor of production, under the interaction of information and technology, has not only made a violent impact on traditional industries and traditional business models, new models and green industrial ecologies with lower energy consumption are constantly emerging. This has not only promoted industrial structural upgrading, but also helped cities transform to low-carbon development. Third, the construction of BDCPZ is biased towards industrial development options. The demand for digital technologies in the pilot zone has spurred the development of high-tech value-added industries, accelerating the rate of

transformation of the industrial structure. In particular, with the increasing maturity of AI technology, it is bound to profoundly change the traditional production and life style and cause changes in the industrial structure (Guo, 2019). This enables the industrial chain to climb to advanced forms of high-value-added, high-tech content, intelligence and green, further promoting the city's shift towards low-carbon and green development.

2.2.2.2. Energy efficiency enhancement effect. Numerous literatures have shown that energy consumption is the main cause of CO₂ emissions (Wang et al., 2011; Acheampong, 2018). Therefore, reducing energy consumption and improving energy use efficiency is the most important and direct path to achieve carbon reduction (Chen, 2009; Tajudeen et al., 2018). The impact of BDCPZ construction on the improvement of urban energy use efficiency is mainly in two aspects. On the one hand, the digital economy can help reduce energy consumption and improve energy efficiency. Data, known as modern oil, has become deeply integrated into every aspect of human production and life in digital age. This is not only reflected in the industrialization of digital but also in the digitization of industry, i.e. the deep penetration of data and digital technology in traditional industries such as agricultural production, processing and manufacturing, and finance. The integration of information technology with other factors of production can more directly contribute to the optimal allocation of production factors in enterprises and reduce energy consumption (Zhang and Wei, 2019). Digital technologies can be used to accurately track all aspects of the industrial chain, including production and sales, in order to optimize the proportion of energy inputs, improve energy efficiency and promote carbon reduction. On the other hand, the construction of BDCPZ promotes the optimization and upgrading of energy production and consumption structure. Digital technologies and big data are increasingly embedded in energy applications and production, facilitating the development of new clean energy sources and accelerating the replacement of traditional energy sources, thus significantly reducing city carbon emissions while improving energy efficiency (Aldieri et al., 2021).

2.2.2.3. Technological progress effect. Green innovation is the most effective path to balance economic growth and environmental protection (Yang et al., 2021), and therefore technological progress is essential for carbon reduction (Acemoglu et al., 2015; Cheng et al., 2018; Yi et al., 2020; Wu et al., 2021b), especially for the manufacturing industry which is the most carbon-emitting sector of all industry. Firstly, Digital empowerment continues to drive green technology innovation (Tang et al., 2021), the application of digital technology has the “multiplier effect” on carbon emission reduction in manufacturing industry. Under the constraints of national “Double Carbon” targets, companies have a greater incentive to innovate green technologies to meet government environmental requirements and social responsibilities. In the short term, a large number of green technology innovations will increase the cost of enterprises and bring a certain “green price”. However, in the long term, the construction of BDCPZ has significantly improved the technological progress of the city's manufacturing industry, especially as data penetrates into all aspects of manufacturing production, the “multiplier effect” is magnified by the increased application of big data (Yang et al., 2022). This process enables enterprises to precisely control the energy consumption of each link and significantly reducing the level of carbon emissions in the manufacturing sector. Secondly, the construction of digital platforms, provide an important intermediary for technological progress and diffusion. When green technology advancement and innovation is developed to a certain extent in the manufacturing sector, it can be transferred to various industries and regions through big data platforms and carriers built in pilot zones. Technology diffusion is conducive to the spread of green technologies to industries and regions with serious environmental pollution (Lin and Zhou, 2021), thus exerting a certain curbing effect on the level of

emissions of society as a whole. Finally, in the era of big data, consumers are pursuing product diversification and the concept of green and sustainable consumption. The changes of consumer consumption perceptions and strong government constraints on environmental regulations have put forward higher standards for enterprises, forcing enterprises to adopt green production methods to meet government and consumer demands.

The specific influence mechanism is shown in Fig. 2. Based on the above analysis, this paper proposes **Hypothesis 2**.

Hypothesis 2. The construction of BDCPZ reduces carbon emission levels in the pilot city by optimizing industrial structure, improving energy use efficiency and promoting GTFP in manufacturing.

3. Research design

3.1. Modeling

China's approval in 2016 to set up the BDCPZ can be seen a quasi-natural experiment. This paper uses the DID model to assess the carbon emission reduction effect of the construction of BDCPZ in China. After excluding some cities with severely missing data samples, we retain the balanced panel data for 261 cities from 2010 to 2019 in this paper. Considering the endogeneity problems associated with omitted variables, we also control year and city and some variables that affect the level of city carbon emissions. The model is as follows.

$$\text{LnCO}_{2it} = \alpha_0 + \beta \text{BDCPZ}_{it} + \delta_i X_{it} + \theta_t + \mu_i + \varepsilon_{it} \quad (1)$$

where LnCO_{2it} is the explanatory variable, denotes the total carbon emissions of city i in year t , BDCPZ_{it} is the core explanatory variable of this paper, expressed the event of construction of BDCPZ, its coefficient β is important, if β is significantly <0 , then it proves that the construction of the BDCPZ has a carbon emission reduction effect. μ_i denotes fixed effect at city level, while θ_t is the time fixed effect and X_{it} is a series of control variables that we selected. In this paper, a parallel trend test is conducted and the PSM-DID method is used to solve the problem of sample self-selection bias.

3.2. Variables and data

3.2.1. Explanatory variables

Total city carbon emissions (logarithmically) are chosen as the explanatory variables in this paper, considering China's “Double Carbon” goal is mainly to reduce total carbon emissions. The specific measurement mainly draws on Wu and Guo (2016) for the idea of measuring total carbon emissions, the Eq. (2) introduces the specific calculation method.

$$\text{CO}_{2it} = \sum_{j=1}^4 \text{CO}_{2ij} = D + E + T + H \quad (2)$$

where CO_{2it} denotes the total carbon emissions of city i in year t , the above formula represents that the carbon emissions measured in this paper contain four components. (1) the carbon emissions from direct energy consumption (D), such as city gas, natural gas and LPG. Emissions have been calculated using the conversion factor provided by the IPCC 2006. (2) carbon emissions generated by the consumption of electrical energy (E) in each city: calculated using the baseline emission factors of each regional power grid and the consumption of electrical energy in the city. (3) carbon emissions from transportation (T) in each city. The energy consumption per unit passenger and freight volume of different modes of transport is calculated by using the various types of energy consumed by the transport sector in *China Statistical Yearbook*, which is combined with the passenger and freight volumes of various modes of transport in the *China City Statistical Yearbook*. (4) Carbon emissions generated from centralized heating (H) in each city. Based on

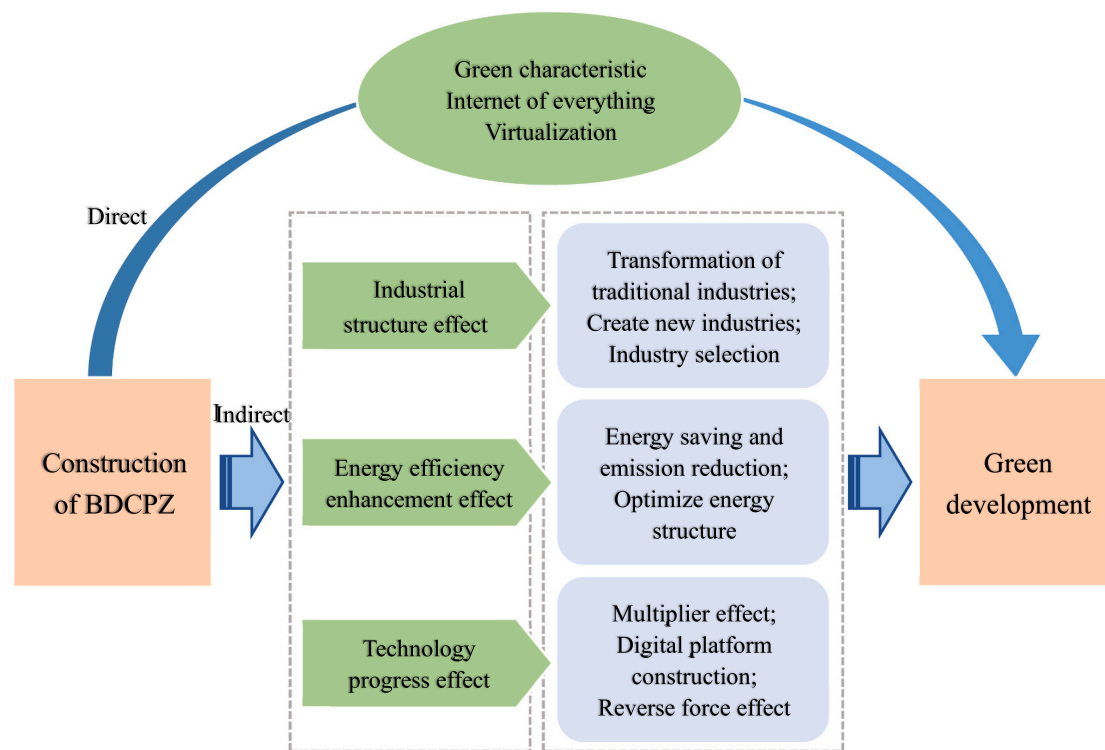


Fig. 2. The influence mechanism of BDCPZ on city green development. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the central heating statistics of each city from the *China Urban Construction Statistical Yearbook*, the raw coal consumption at the city level was inferred and then calculated using the raw coal consumption. Finally, the carbon emissions of the above four components are summed to obtain the total carbon emissions of the city.

3.2.2. Core explanatory variables

The establishment of the BDCPZ. The variable *Treat* is a dummy variable of the treatment group, the method of selection of treatment groups is detailed in Section 2.1. The variable *Post* is a dummy variable of time, which indicates the establishment of the BDCPZ. If the sample period belongs to 2016 and later, the value is 1; otherwise, the value is 0. Multiply *treat* with *post* to obtain the policy dummy variable *BDCPZ*.

3.2.3. Control variables

This paper controls for additional factors influencing total city carbon emissions: population density (*Pop_density*). High population density represents economic agglomeration, which may increase carbon emissions, but the relative concentration of factors may also increase the efficiency of energy and thus reduce carbon emissions. Level of government intervention (*Gov*) expressed as the ratio of local budget expenditure to GDP. The environmental regulatory tools used by governments may have some impact on the emission reduction behavior of emitters. The level of financial development (*Fin*) is expressed as the ratio of year-end loan balances of financial institutions to GDP. Financial development promotes economic growth and thus increases carbon emissions, but financial development also has the potential to promote technological progress and increase resource efficiency to reduce carbon emissions. The size of the local economy (*LnGDP*). The level of regional economic development is closely related to carbon emissions. The level of openness to the outside world (*Open*) is expressed as the ratio of total actual foreign capital utilized to GDP. Foreign capital inflows may increase productivity and reduce carbon emissions, but they may also lead to “pollution havens” that increase production and energy consumption. The strength of investment in science and education (*RD*) is expressed as

the ratio of the sum of local expenditure on science and education to local budgetary expenditure. Technological progress is a key driver of the green transition, and investment in science and education is a source of technological progress and innovation. Employment structure optimization (*Jobs*) plays an irreplaceable role in greening and upgrading industries, and we control for this variable, expressed as the ratio of employment in productive services to employment in manufacturing.

3.2.4. Data sources

All data in this paper were obtained from the *China City Statistical Yearbook*, *China Energy Statistical Yearbook*, statistical yearbooks of various cities, as well as the *Information Network of the Development Research Centre of the State Council and the China, Stock Market & Accounting Research Database*. Some missing data was added by consulting statistical bulletins on national and city economic and social development. The descriptive statistics are shown in Table 1.

Table 1
Descriptive statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max
LnCO ₂	2610	6.3499	1.1221	3.0927	9.5330
BDCPZ	2610	0.0598	0.2371	0	1
Pop_density	2610	5.8156	0.8795	1.6198	7.9227
Gov	2610	0.1848	0.0796	0.0439	0.6750
Fin	2610	0.9549	0.5667	0.1180	5.3047
LnGDP	2610	16.5800	0.9102	12.6455	19.6803
Open	2610	0.0180	0.0184	0.0000	0.2101
RD	2610	0.1942	0.0425	0.0055	0.3722
Jobs	2610	0.8646	1.0102	0.0556	16.0651
Ind	2610	0.9476	0.5189	0.1756	5.1543
E.power	2610	12.0682	0.7097	7.4879	14.7210
GTFP	2610	0.9834	0.0528	0.6960	1.5920

4. Empirical results and analysis

4.1. Baseline regression

It has become a general consensus that big data can play a positive role in promoting productivity improvement and social transformation and will undoubtedly contribute to the achievement of China's "Double Carbon" goal. Column (1) and column (2) in Table 2 take the total carbon emissions as the explained variable. These results indicate that the construction of the BDCPZ significantly reduced the level of carbon emissions in the city. In terms of the economic significance of the estimated coefficients, the BDCPZ significantly reduced the total carbon emissions of cities in pilot zones by 29.6% compared to cities that are not part of the BDCPZ, suggesting that the construction of the BDCPZ could be an essential force in achieving the "Double Carbon" goal for Chinese society in the future. The above findings validate **Hypothesis 1**.

4.2. Robustness tests

4.2.1. Substitution of explanatory variables

Among the indicators for measuring regional carbon emissions, per capita carbon emissions focus more on the measure of equity, while carbon intensity focuses more on the measure of efficiency (Wu and Guo, 2016). This paper also takes into account both efficiency and equity, two important perspectives of concern to society, by replacing the explanatory variables with carbon emissions per capita and carbon emission intensity. The specific regression results are shown in columns (3) and (4) in Table 2, the basic conclusion of this paper is still valid. We also used data from the CEADs database on city carbon emissions in China for robustness tests. The specific results are shown in Table 3.

4.2.2. Parallel trend test

The validity of the double difference approach to policy assessment is based on the assumption of parallel trends. Drawing on the treatment of Beck et al. (2010), this paper further constructs Eq. (3).

$$\text{LnCO}_{2it} = \alpha_0 + \beta_k \sum_{k=-6}^{k=3} D_{it}^k + \gamma M_{it} + \delta_j X_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (3)$$

where D_{it}^k is a series of dummy variables indicating the event of the construction of BDCPZ. The data range for this paper includes the years

Table 2
Baseline model results.

Variables	LnCO ₂	LnCO ₂	CO ₂ /POP	CO ₂ /GDP
	(1)	(2)	(3)	(4)
BDCPZ	−0.2638*** (0.0346)	−0.2963*** (0.0350)	−0.7590*** (0.1172)	−0.1545*** (0.0319)
Pop.density		0.6297*** (0.1936)	−2.7272** (1.3197)	1.4425*** (0.4169)
Gov		−0.0357 (0.3363)	−1.7991 (1.1897)	−2.1743*** (0.8134)
Fin		−0.1341** (0.0653)	0.8888** (0.4496)	−0.2905* (0.1653)
LnGDP		0.0075 (0.0315)	−0.4603** (0.1882)	−2.4312*** (0.6013)
Open		1.6494*** (0.5606)	−1.2133 (2.2094)	2.5950** (1.2903)
RD		−0.8379** (0.3883)	1.2950 (1.4562)	−0.6557 (0.4672)
Jobes		−0.0504*** (0.0125)	−0.0042 (0.0380)	−0.0103 (0.0110)
City	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	2610	2610	2610	2610
R ²	0.9132	0.9153	0.8547	0.6586

Note: Heteroskedasticity robust standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3

Robustness test: change the measurement method (from CEADs).

Variables	LnCO ₂	LnCO ₂
	(1)	(2)
BDCPZ	−0.1129*** (0.0212)	−0.1201*** (0.0221)
Control	No	Yes
City	Yes	Yes
Year	Yes	Yes
N	2027	2027
R ²	0.9613	0.9616

2010–2019, covering six years prior and three years after the implementation of the BDCPZ. This paper uses 2010 as the base period and removes that dummy variable as a control in the regression. This section focuses on the parameter β_k .

In this paper, the regression coefficients and 95% confidence intervals for the parameter β_k in Eq. (3) are plotted. As can be seen from Fig. 3, the value of β_k was relatively flat and not significantly different from zero before the construction of the BDCPZ was implemented. This indicates the sample we selected satisfies the assumption of parallel trend. In terms of dynamic effects, the estimated value of the parameter β_k in the carbon emission equation was always negative and increased in absolute value from the first to the third year of the construction of the BDCPZ. This indicates that the inhibitory effect is significant and increases year by year during the sample period selected for this paper.

4.2.3. Placebo test

To ensure the carbon reduction effect in the baseline estimates was indeed caused by the construction of the BDCPZ, a placebo test was conducted. Specifically, we randomly select cities for policy implementation, and the experiment was repeated 500 times. The estimated coefficients for the virtual policy implementation group are distributed around 0 and obey a normal distribution (See Fig. 4), which is far from the estimated coefficient in the benchmark regression results (−0.2963), i.e. the constructed virtual treatment effect does not exist.

4.2.4. PSM-DID results

The PSM method was used to remove the sample self-selection bias prior to the implementation of the DID, in order to assess accurately. The regression results in Table 4 show that the level of carbon emissions in the pilot cities decreased significantly after the big data pilot zones were built, a finding that remains robust after overcoming sample self-selection bias.

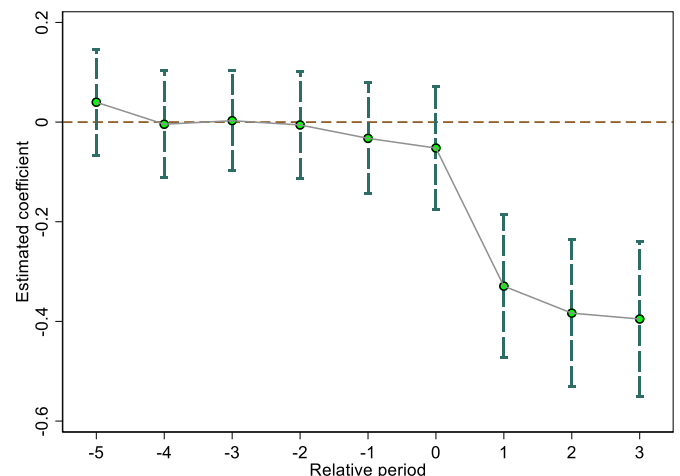


Fig. 3. Parallel trend test.

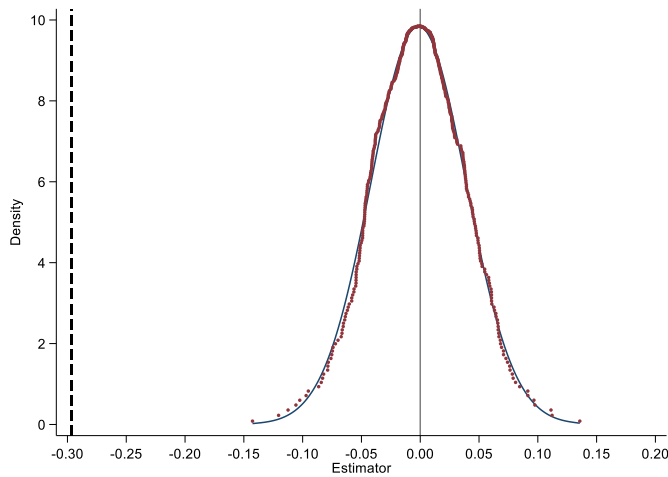


Fig. 4. Placebo test.

Table 4
Robustness test: PSM-DID.

Variables	LnCO ₂ (1)	LnCO ₂ (2)
BDCPZ	-0.2485** (0.1141)	-0.3235*** (0.1120)
Control	No	Yes
City	Yes	Yes
Year	Yes	Yes
N	248	248
R ²	0.9550	0.9605

4.2.5. Excluding the impact from other policies

While the construction of the BDCPZ released carbon emission reduction dividends to the city, it may also be affected by a variety of other policies before and after the construction of the pilot zone. Here, the three main policy shocks during the sample period are excluded: Carbon emissions trading pilot, Low-Carbon City Pilot and National Innovative City Pilot Policy. Results in Table 5 show that the magnitude and direction of the coefficients of the BDCPZ did not change significantly, and it can be argued that other policies affecting carbon reduction during the same period do not substantially interfere with the baseline regression results in this paper.

4.2.6. Replacing the control group

This paper further employs the replacement of control groups to test the baseline regression results. Specifically, only other cities in the province where the pilot city is located are included in the control group. Results in Table 6 show the conclusion that the construction of BDCPZ has a significant carbon reduction effect still holds when the cities in the

Table 5
Robustness tests: excluding the impact other policy.

Variables	LnCO ₂ (1)	LnCO ₂ (2)	LnCO ₂ (3)
BDCPZ	-0.2769*** (0.0355)	-0.3054*** (0.0343)	-0.2745*** (0.0334)
Carbon Trading	Yes		
Low Carbon		Yes	
Innovative City			Yes
Control	Yes	Yes	Yes
City	Yes	Yes	Yes
Year	Yes	Yes	Yes
N	2610	2610	2610
R ²	0.9165	0.9158	0.9185

Table 6
Robustness tests: Replacing the control group.

Variables	LnCO ₂ (1)	LnCO ₂ (2)
BDCPZ	-0.2828*** (0.0572)	-0.1762*** (0.0650)
Control	No	Yes
City	Yes	Yes
Year	Yes	Yes
N	780	780
R ²	0.9107	0.9182

pilot area are included in the control group with other cities in the same province.

4.3. Mechanism of action

Both the baseline regression model and robustness tests in the previous section confirm that the construction of BDCPZ in China has a significant carbon reduction effect. How is this effect achieved? According to the theoretical analysis in Section 2.2, the possible mechanisms including optimizing industrial structure, improving energy efficiency, and promoting green TFP in manufacturing. This section examines both mechanisms using Eqs. (4) and (5).

$$M_{it} = \lambda_0 + \lambda_1 BDCPZ_{it} + \lambda_2 X_{it} + \theta_t + \mu_i + \varepsilon_{it} \quad (4)$$

$$LnCO_{2it} = \alpha_0 + \beta BDCPZ_{it} + \gamma M_{it} + \delta_j X_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (5)$$

M is the mechanism variable. Among them, the level of advanced industrial structure (*Ind*) is expressed as the ratio of added value between the tertiary industry and the secondary industry, energy use efficiency (*E_{power}*) is expressed as the ratio of GDP to total energy consumption (calculated in standard coal) of the city, and green total factor productivity (*GTFP*) of the manufacturing industry is measured by SBM-ML model.

Column (1) in Table 7 show that the construction of BDCPZ significantly Optimized city's industrial structure. Column (2) shows that industrial structure optimization is a powerful mechanism to reduce city carbon emissions, which is consistent with existing studies (Zhao et al., 2022; Wu et al., 2021a; Wang et al., 2019b). The construction of big data promotes technological upgrading and productivity improvement within traditional industries, and also continues to give rise to smart, green new industries. Therefore, BDCPZ construction has certain industrial structure upgrading effect to promote carbon emission reduction.

As can be seen from columns (3) and (4) in Table 7, the construction of BDCPZ also promotes energy efficiency, and the regression coefficients of energy efficiency is significant, which is consistent with existing studies (Yang et al., 2020; Li et al., 2021; Zhang et al., 2022d). The empirical results prove that the construction of BDCPZ can effectively lead to a change in the regional economic growth model from a crude development model with high energy consumption and emissions to an intensive and green development model, make further efforts to enhance energy efficiency and promote green development of the city. Therefore, improving energy efficiency is one of the impact mechanisms for the BDCPZ construction to release a dividend of carbon reduction.

From columns (5) and (6) in Table 7, the construction of BDCPZ can promote GTFP, and the two together reduce carbon emissions. The construction of BDCPZ has brought about more biased technological advances and a greater focus on greening and energy conservation, and green technological progress is an important force for carbon reduction (Dong et al., 2022; Du et al., 2019; Lin and Ma, 2022). At the same time, as the manufacturing industry is already an essential source of carbon emissions, the application of digital technologies promotes green technological innovation by accurately controlling energy consumption,

Table 7
Results of mechanisms tests.

Variables	<i>Ind</i>	<i>LnCO₂</i>	<i>E_{power}</i>	<i>LnCO₂</i>	<i>GTFP</i>	<i>LnCO₂</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>BDCPZ</i>	0.0314* (0.0171)	−0.2911*** (0.0349)	0.0592** (0.0242)	−0.2583*** (0.0330)	0.0273*** (0.0072)	−0.2716*** (0.0351)
<i>Ind</i>		−0.1647*** (0.0528)				
<i>E_{power}</i>				−0.6402*** (0.0298)		
<i>GTFP</i>						−0.9062*** (0.1776)
<i>Control</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>City</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	2610	2610	2610	2610	2610	2610
<i>R²</i>	0.9192	0.9157	0.8723	0.9368	0.1765	0.9168

releasing the “multiplier effect” and building platforms, thereby increasing the GTFP of manufacturing industry and reducing carbon emissions. Therefore, the improvement of manufacturing GTFP is one of the mechanisms. The above findings validate **Hypothesis 2**.

4.4. Heterogeneity analysis

The effect may vary in numerous ways due to differences in the endowments of individual cities and inherent characteristics of the construction of BDCPZ. In this paper, heterogeneity is analyzed from five perspectives: the size of the city's population, the level of development of its manufacturing sector, its resource endowment, and whether it has been selected as a pilot carbon trading right and a pilot low-carbon city.

4.4.1. Population size of the city

Population size of the city is an important factor affecting carbon emission (Shi et al., 2023). We divided cities into two groups with a population of 5 million as the boundary, columns (1) and (2) in Table 8 show that the larger the population of a city, the stronger the carbon reduction effect of BDCPZ construction. The reason may lie in the fact that larger cities typically enjoy higher levels of resource allocation, circulation, and utilization efficiency, as well as stronger advantages in terms of economic and resource agglomeration. The construction of BDCPZ utilizes big data and information technology to transform the city development model, revolutionize the city governance model, improve city resource allocation and utilization efficiency, and contribute to the reduction of carbon emissions. As cities transform their development models, modify their governance models, and especially deepen their digital technology innovation and application, the “city disease” in the traditional urban development process can be alleviated to a certain extent, which provides a novel path for exploring development models in the future.

Table 8
Analysis of heterogeneity (I).

Variables	Population size		Manufacturing development		Resource endowment	
	Small City	Big City	Advanced Manufacturing Base	Non-Advanced Manufacturing Base	Resource-Based City	Non-Resource-Based City
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>LnCO₂</i>	<i>LnCO₂</i>	<i>LnCO₂</i>	<i>LnCO₂</i>	<i>LnCO₂</i>	<i>LnCO₂</i>
<i>BDCPZ</i>	−0.2447*** (0.0491)	−0.3545*** (0.0494)	−0.3365*** (0.0477)	−0.1889*** (0.0407)	−0.1456* (0.0847)	−0.3025*** (0.0424)
<i>Control</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>City</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1625	979	522	2088	612	1998
<i>R²</i>	0.8856	0.9469	0.9460	0.8807	0.8716	0.9408

4.4.2. Level of manufacturing development

The manufacturing sector is the second largest source of greenhouse gas emissions in China, and its success in reducing emissions has a direct impact on the realization of the “Double Carbon” goal. In this paper, we divide the samples into two groups according to the 2020 China Advanced Manufacturing City Development Index released by CCID Consulting Company Limited. A city is assigned a value of 1 if it is among the top 50 cities in the country in terms of advanced manufacturing development index, and a value of 0 otherwise. The results of the sub-sample regression show that the carbon emission reduction effect of the construction of BDCPZ is significant regardless of whether it is an advanced manufacturing base or not, but the effect is better released in the advanced manufacturing base.

The reason may be that the construction of BDCPZ has strengthened the application of big data and digital technology in various processes of manufacturing production, which can achieve more accurate control of energy consumption in various processes, reduce wasteful redundancy, and provide important technical support and data support for energy conservation and emission reduction. In addition, there may be a certain scale effect in the construction of the Big Data Comprehensive Pilot Zone, which will have a greater inhibiting effect on carbon emissions in cities with a higher level of manufacturing development.

4.4.3. Resource endowment

The carbon emission reduction dividend of the construction of BDCPZ may vary depending on the city's resource endowment. Columns (5) and (6) in Table 8 show that the carbon emission reduction effect of the BDCPZ on non-resource-based cities is greater. This might be because in the process of promoting industrialization and urbanization, resource-based cities are prone to form a solidified development model that relies on their resource endowments. Resource-rich cities may face the “resource curse” when promoting green transformation and digitalization (Li and Xu, 2018).

4.4.4. Carbon emissions trading pilot

As an effective policy tool, carbon emissions trading can facilitate the flow of resources and increase resource efficiency. In part, this policy forces high-carbon firms to reduce emissions by introducing new technologies, while providing incentives for low-carbon firms to continue to develop new technologies to maintain their advantage in order to expand their market share. Therefore, the effect will be further enhanced in cities with pilot carbon emissions trading. Columns (1) and (2) in Table 9 verify the above findings. This policy and the construction of BDCPZ complement each other and provide some guidance for the government to develop a diversified carbon emissions governance system in the future.

4.4.5. Low-carbon city pilot

We extend the heterogeneity analysis based on whether a city is included in a low-carbon city pilot. The regression results in columns (3) and (4) in Table 9 show the effect is greater for low-carbon cities, suggesting that this policy also strengthens the carbon reduction from BDCPZ construction. The reason may be that the low-carbon city pilot policy is an environmentally binding policy (Zhuang, 2020), and government environmental regulation can induce green technology innovation in firms (Xu and Cui, 2020). The previous analysis shows that the digital economy can also facilitate green technology innovation by firms. A combination of the two policies would yield more desirable carbon reductions.

5. Conclusion and policy implications

5.1. Conclusions

Leveraging the power of data to promote carbon emission reduction is an important tool to achieve the “Double Carbon” goals. This paper examines the power of big data to reduce carbon emissions and its specific mechanisms using DID model with a panel of 261 cities in China from 2010 to 2019. The findings are as follows: First, the construction of BDCPZ significantly reduced the carbon emissions of the cities in the pilot zone. The finding is robust. Secondly, the mechanism analysis shows that there are three mechanisms for the impact of the implementation of the BDCPZ on the level of city carbon emissions: optimizing industrial structure, improving energy efficiency and promoting GTFP in manufacturing. Third, the carbon emission reduction effect of BDCPZ varies depending on city population size, levels of manufacturing development, resource endowments, and pilot cities with low carbon related policies. The combination of some policies is more helpful in releasing the carbon reduction dividend of BDCPZ construction.

5.2. Policy implications

First, it is important for Chinese governments at all levels to strengthen top-level design for big data and the digital economy. Timely summaries to construct empirical and reproducible models should be developed. Provinces and cities that have established pilot zones should continue to firmly implement their big data development strategies. Provinces that have not established big data pilot zones should fully consider their own realities, keep up with the technological changes brought by big data and deploy big data development strategies at the earliest.

Second, it is essential to take full advantage of the effects of industrial structure optimization, efficiency enhancement and green technology advancement to build comprehensive pilot zones for future big data. Investing aggressively in the construction of information technology infrastructure and properly increasing investment in big data, 5G and artificial intelligence. Based on technological advancement, the city will optimize and upgrade its industrial structure. Further deep integration of digital technologies such as big data with cities, industries and enterprises will be accelerated, and the penetration of big data into all

Table 9

Analysis of heterogeneity (II).

	Carbon Emissions Trading Pilot	Non-Carbon Emissions Trading Pilot	Low-Carbon City Pilot	Non-Low-Carbon City Pilot
	(1)	(2)	(3)	(4)
Variables	LnCO ₂	LnCO ₂	LnCO ₂	LnCO ₂
BDCPZ	−0.3649*** (0.0863)	−0.1940*** (0.0389)	−0.3692*** (0.0550)	−0.1864*** (0.0442)
Control	Yes	Yes	Yes	Yes
City	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	245	2365	978	1549
R ²	0.9430	0.9152	0.9294	0.9102

aspects of production and life, especially into traditional industries, will be promoted. Strengthen the application of digital technology in all areas of production and life, transportation and urban governance.

Third, it is necessary to implement differentiated strategies for the big data to avoid a “one-size-fits-all” policy. For cities with large populations, it is important to vigorously develop big data-related technologies, focus on enhancing synergies between data and other factors of production, and mitigate the back-reaction of population expansion on green development. For cities with high levels of manufacturing development, digital technology should be applied to all production processes to more precisely control energy consumption and reduce carbon emissions. Resource-based cities should seize new opportunities in the development of the digital economy, actively develop digital technologies, enrich regional industries and strengthen the carbon reduction power of big data development. Moreover, the government should strengthen the “green” orientation of the construction of BDCPZ, pay more attention to the synergy between different policies, and eventually form a policy synergy that leads and matches each other, so as to play a good policy “combination fist”.

5.3. Limitations

The limitations of this paper mainly focus on the following two aspects. On the one hand, due to the lack of uniformity in the measurement of carbon emissions, and the fact that data on carbon emissions from Chinese cities in databases such as CEADs are only updated to 2019, this paper only revealed the short-term impact of BDCPZ policy on city carbon reduction. In the future, more long-term panel data are needed to characterize the digital dividend and dynamic influence of BDCPZ. On the other hand, this study only focuses on the impact of BDCPZ policy on city green development. However, the construction of big data should have broader social impacts, including employment, innovation efficiency and even income distribution etc., and future research should focus more on these aspects.

CRedit authorship contribution statement

Xiahai Wei: Conceptualization, Methodology, Software, Data curation, Writing – original draft. **Feng Jiang:** Conceptualization, Methodology, Data curation, Writing – original draft. **Lisha Yang:** Conceptualization, Methodology, Data curation, Writing – original draft.

Declaration of Competing Interest

The authors have declared no conflict of interest.

Data availability

Data will be made available on request.

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